Start with the two racks

My project described a physical system. The emulator exercise gave me a software controller to load that project onto.

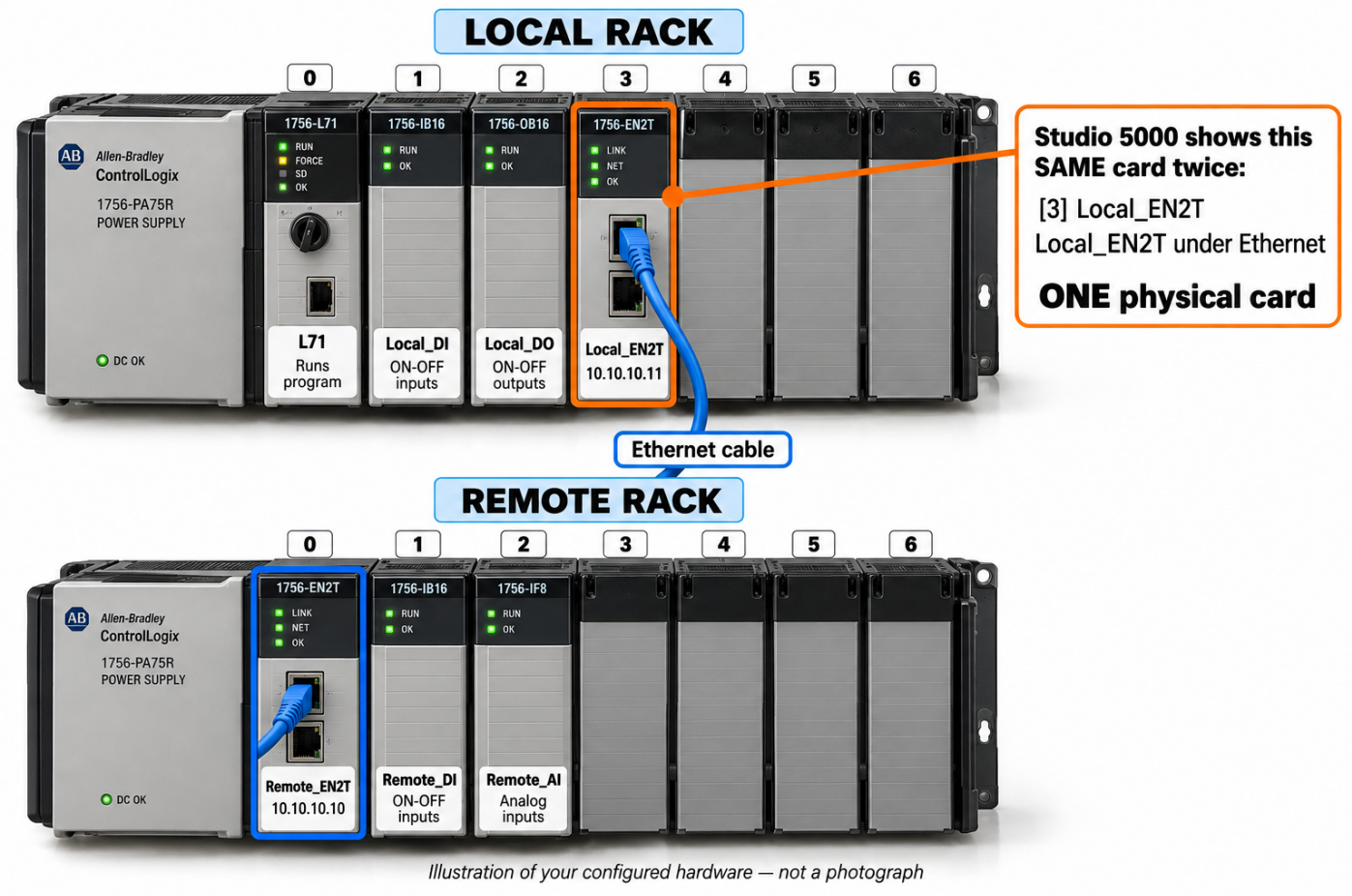


Illustration of your configured hardware — not a photograph

Follow the remote input signal

The remote input card passes its data across the remote backplane to Remote_EN2T in remote slot 0. The data crosses Ethernet to Local_EN2T in local slot 3, then crosses the local backplane to the controller in local slot 0.

Read the picture as a layout

A rack holds the modules. A slot is a numbered position. A backplane connects modules inside a rack. The two Ethernet cards connect the racks. Each rack has its own slot 0.

The project still has the input and output layout

Local slots 1 and 2 hold the discrete input and output modules. Remote slots 1 and 2 hold the discrete and analog input modules. The picture is a layout illustration. The configured local output module was 1756-OB16E.

Trace the software connection

Imagine the programming laptop and the controller in a panel. In this exercise the panel and controller live in software on the same computer.

1 Logix Designer

I open the project file here. This is where I edit the program and choose Download, Upload, or Go Online.

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2 FactoryTalk Linx

The communication software my Logix Designer session used to exchange messages with the controller.

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3 Virtual backplane

The software connection inside the simulated rack. This is the branch called Backplane in my browser.

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4 Emulate 5570 in slot 0

The simulated controller. It receives the download and executes the project in Run mode.

Read the address like a route

Backplane\0 means go through the virtual backplane to the controller in slot 0. In Who Active, I expand Backplane and select the emulator row labeled 00. Selecting the branch itself only points to the rack.

Where the physical picture fits

The simulated controller takes the role of the controller in local slot 0. The Ethernet cards in the picture belong to the separate path between the controller and the remote input modules.

Which screen does what?

A driver handles a kind of connection. A browser shows what it can reach. A project setting tells Logix Designer what controller the file is built for.

My route / FactoryTalk Linx

In Logix Designer, open **Who Active**. Expand **Backplane** and select **00, Studio 5000 Logix Emulate**. My selected path was **Backplane\0**.

Lesson route / RSLinx Classic

Open **RSWho** in RSLinx Classic to browse. Expand **AB_VBP-1**, the virtual backplane driver. The teacher also browsed this route in **Who Active** and selected his emulator in slot 1.

What does AB_VBP-1 mean?

VBP stands for virtual backplane. The full string is the driver name shown by RSLinx Classic. The slot number comes from the emulator row underneath it. On the teacher's screen, the path ended in **\1** because he selected slot 1.

What would I add in Configure Drivers?

A communication route for the connection I intend to use. My FactoryTalk Linx Backplane already showed the emulator. That was enough to select the destination in my Who Active window.

What would I change in Controller Properties?

The project target. I could see the emulator in the browser while the project still said 1756-L71. I changed the project to Emulate 5570, revision 34.011, slot 0. The teacher used revision 24.11 and slot 1 in his separate project.

Why the teacher changed to 1756-L75

His Change Controller dialog showed a move from an emulator to a physical controller type. My exercise went from a physical controller type to an emulator so I could run my assignment in software.

Why COM3 came up

Trace the attempted serial route and the route that actually carried the download.

Serial route / Driver setup

RSLinx Classic tries to open the computer's serial port.

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COM3 / Port could not open

The attempt stopped at the computer, before it could send a message down a serial connection.

What would come next?

If the port opened, the driver could send a message through a serial cable or adapter and wait for a reply.

My route / Who Active

FactoryTalk Linx browses the virtual rack on the computer.

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Backplane / Working route

The virtual rack provides a software path to the controller.

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Slot 0 / Emulator

The project downloaded here. The transfer reported zero errors.

What each serial label tells me

RS-232 identifies the serial interface. **DF1** is the protocol, or the rules for the messages. **COM3** is the Windows name the driver was told to open for its end of that connection.

What the error actually proves

RSLinx Classic could not open the selected port. The message does not identify the cause or establish whether a controller was on the far end. A missing or busy port could produce that symptom.

Why I could keep going

The emulator was reachable inside my computer through the virtual rack. That path has no COM3 step. The DF1 channel shown under the emulator was a browse entry, while the selected communication path was Backplane\0.

Run the exercise in order

At each step, ask what I just verified before moving to the next one.

1. Find the controller

In Who Active, expand Backplane. Seeing the emulator in slot 00 verifies that the communication software can find it.

2. Check what my project is built for

Controller Properties should show Emulate 5570, revision 34.011, slot 0 for this exercise. The browser and project properties answer different questions.

3. Send the project to the controller

Select the emulator row at Backplane\0 and choose Download. The first attempt stopped at an empty ladder rung. Removing that rung let the download finish with zero errors.

4. Check controller status

Rem Run means Remote Run. Controller OK and input/output not responding can appear together because the software controller is running while the configured physical modules are absent. Simulating input values for logic testing is a separate exercise.

5. Read back or connect live

Closing the project showed No Controller in the editor. I selected the emulator again and opened Connected To Upload. The dialog found my matching saved project file. Upload reads the controller into Logix Designer. Go Online connects live using the matching file. My screenshot confirms the dialog stage.

What I want to review

I want to trace this like a wire on a drawing. Where am I starting? What carries the data? Which controller is on the other end? Does the project match? Am I sending the project, reading it back, or just connecting?